Appendix A. The Hardy Z-Function as a Weakly Harmonizable Time-Warped Stationary Gaussian Process

This appendix develops, in self-contained form, the second-order stochastic classification of the Hardy Z-function under the spectral representation derived in the main text. We prove that Z is a centered Gaussian process of the weakly harmonizable, time-warped stationary class (equivalently $G(\theta)$-stationary, reducibly-to-weakly-stationary), derive its exact autocovariance as a convergent spectral integral, extract the variance structure function from the diagonal, and verify that the resulting variance is logarithmic with an explicit uniform error constant of $\pm\frac{1}{100}$.

A.1. **Setting and notation.** Throughout this appendix we fix $T_0 \geq 200$ and write

$$\theta(t) \;=\; \operatorname{Im}\log\Gamma\big(\tfrac14+\tfrac{it}{2}\big)-\tfrac{t}{2}\log\pi, \tag{1}$$

$$\theta'(t) \;=\; \tfrac12\operatorname{Re}\psi\big(\tfrac14+\tfrac{it}{2}\big)-\tfrac12\log\pi, \tag{2}$$

$$Z(t) \;=\; e^{i\theta(t)}\zeta\big(\tfrac12+it\big)\,, \tag{3}$$

the Riemann–Siegel theta function, its derivative, and the Hardy Z-function, respectively. By the Binet first formula applied to $\frac12\operatorname{Re}\psi(\frac14+\frac{it}{2})$, one has the uniform bound

$$\Big|\theta'(t)-\tfrac12\log(t/(2\pi))\Big| \;\leq\; \tfrac{1}{200} \qquad (t\geq 200), \tag{4}$$

with $\theta''(t)>0$ on $[T_0,\infty)$ so that θ is a C^∞-diffeomorphism onto $[\theta(T_0),\infty)$.

Let H denote the process on the θ-axis defined by

$$H(u) \;:=\; \frac{Z(\theta^{-1}(u))}{\sqrt{\theta'(\theta^{-1}(u))}}, \qquad u\in[\theta(T_0),\infty). \tag{5}$$

Equivalently,

$$Z(t) \;=\; \sqrt{\theta'(t)}\;H(\theta(t)). \tag{6}$$

We assume throughout that H admits a Cramér spectral representation

$$H(u) \;=\; \int_{-1}^{1} e^{i\xi u}\,d\Phi(\xi), \qquad \mathbb{E}\Big[d\Phi(\xi)\overline{d\Phi(\eta)}\Big] \;=\; \delta(\xi-\eta)\,S(\xi)\,d\xi, \tag{7}$$

with orthogonal complex Gaussian random spectral measure $d\Phi$ and spectral density $S(\xi)\geq 0$ supported in $[-1,1]$. The support claim is the content of the spectral-tiling and remainder-atom theorem of the main text, and the normalization

$$\int_{-1}^{1} S(\xi)\,d\xi \;=\; 2 \tag{8}$$

is pinned by the Titchmarsh second-moment formula [7].

A.2. **Classification theorem.**

Theorem A.1 (Classification)**.** *The process Z defined by* (3) *is a centered Gaussian process on* $[T_0,\infty)$ *belonging to the following classes simultaneously:*

1. Reducibly weakly stationary *(RWS): $H = Z\circ\theta^{-1}/\sqrt{\theta'\circ\theta^{-1}}$ is weakly stationary on* $[\theta(T_0),\infty)$ *(*[3, Ch. 4]*,* [6]*).*
2. $G(\theta)$-stationary *in the sense of Gray, Vijverberg, and Woodward: Z admits the generalized spectral representation*

$$Z(t) \;=\; \sqrt{\theta'(t)}\int_{-1}^{1} e^{i\xi\theta(t)}\,d\Phi(\xi) \tag{9}$$

against the deformed harmonic kernel $e^{i\xi\theta(t)}$ ([6]*).*

(3) Deformed (time-warped) stationary *in the sense of Clerc–Mallat:* $Z = D_\theta H$*, where* $(D_\theta f)(t) := \sqrt{\theta'(t)}\, f(\theta(t))$ *is a unitary operator* $L^2(\mathbb{R}) \to L^2(\mathbb{R})$ *(*[5]*).*

(4) Weakly harmonizable *in the sense of Rozanov and Rao:* Z *has a spectral bimeasure of finite Fréchet variation and infinite Vitali variation, since* $\operatorname{Var} Z(t) \to \infty$ *as* $t \to \infty$ *(*[4], [2]*).*

Z *is* not *an oscillatory process in the sense of Priestley and* not *an oscillatory harmonizable process in the sense of Chang–Rao.*

Proof. Gaussianity. The spectral random measure $d\Phi$ in (7) is complex Gaussian, so $H(u)$ is a centered Gaussian process. Since $Z(t)$ is a deterministic multiple of H at a deterministic argument, it is also centered Gaussian.

(1) RWS. From (5), setting $u = \theta(t)$ gives $H(u) = Z(\theta^{-1}(u))/\sqrt{\theta'(\theta^{-1}(u))}$, whose covariance depends on (u_1, u_2) only through $u_1 - u_2$ by (7). Thus H is weakly stationary and θ is the reducing diffeomorphism.

(2) $G(\theta)$-stationary. Substituting (7) into (6) yields (9), the defining form of $G(\theta)$-stationarity.

(3) Deformed stationary. The operator D_θ is unitary on $L^2(\mathbb{R})$: for any $f \in L^2$,

$$\int |(D_\theta f)(t)|^2\, dt = \int \theta'(t) |f(\theta(t))|^2\, dt = \int |f(u)|^2\, du,$$

by the change of variable $u = \theta(t)$, $du = \theta'(t)\, dt$. Thus $Z = D_\theta H$ represents Z as the image of a weakly stationary process under a unitary stationarity-breaking operator.

(4) Weak harmonizability. By Theorem A.4 below, $\operatorname{Var} Z(t) = \log(t/(2\pi)) \pm \frac{1}{100}$, which is unbounded as $t \to \infty$. Therefore the spectral bimeasure of Z has infinite Vitali (total) variation, so Z is not strongly harmonizable. Its bimeasure, however, is the pushforward of the orthogonal measure $d\Phi$ under the diffeomorphism θ, which has finite Fréchet variation by (8). Hence Z is weakly harmonizable.

Exclusion from Priestley and Chang–Rao classes. Write (9) formally against the rigid harmonic kernel $e^{i\lambda t}$ with $\lambda = \xi$, giving amplitude envelope

$$A_t(\lambda) \;=\; \sqrt{\theta'(t)}\, e^{i\lambda[\theta(t) - t]}.$$

Priestley's definition requires the t-Fourier transform $|\widehat{A_\bullet(\lambda)}(\theta)|$ to be concentrated at $\theta = 0$; Chang–Rao drop the concentration but require the transform to attain a maximum somewhere on $\mathbb{R}$. With total phase $\Phi(t; \theta, \lambda) = \lambda[\theta(t) - t] - \theta t$, the stationary point satisfies $\theta'(t^*) = 1 + \theta/\lambda$, i.e., $t^*(\theta, \lambda) = 2\pi e^{2 + 2\theta/\lambda}$ by (4). With $\Phi''(t^*) = \lambda/(2t^*)$, the leading amplitude of the stationary-phase contribution is

$$|\widehat{A_\bullet(\lambda)}(\theta)| \;\asymp\; 2\pi \sqrt{2/|\lambda|}\, e^{1 + \theta/\lambda} \sqrt{1 + \theta/\lambda},$$

which is strictly increasing and unbounded as $\theta/\lambda \to +\infty$. Hence $A_t(\lambda)$ has no Fourier maximum on $\mathbb{R}$ and is not an oscillatory function in either sense. ☐

A.3. **Exact autocovariance.**

Theorem A.2 (Exact autocovariance)**.** *For all* $t, s \in [T_0, \infty)$*,*

$$R_Z(t, s) \;:=\; \mathbb{E}[Z(t)\overline{Z(s)}] \;=\; \sqrt{\theta'(t)\theta'(s)} \int_{-1}^{1} e^{i\xi[\theta(t) - \theta(s)]}\, S(\xi)\, d\xi. \tag{10}$$

In particular, R_Z *depends on* (t, s) *only through the pair* $(\theta(t) - \theta(s),\, \theta'(t)\theta'(s))$*.*

Proof. From (6) and (7),

$$\mathbb{E}[Z(t)\overline{Z(s)}] = \sqrt{\theta'(t)\theta'(s)}\, \mathbb{E}\left[\int e^{i\xi\theta(t)} d\Phi(\xi) \int e^{-i\eta\theta(s)} \overline{d\Phi(\eta)} \right].$$

By the orthogonality relation in (7), $\mathbb{E}[d\Phi(\xi)\overline{d\Phi(\eta)}] = \delta(\xi - \eta) S(\xi) d\xi$, so the double integral collapses to (10). ☐

Corollary A.3 (Sample-path inner-product form)**.** *For any window* $[T_0, T]$ *and any lag* η*, the sample-path autocorrelation of* H *is*

$$C_H(\eta) \;=\; \int_{T_0}^{T_\eta} Z(\tau)\, Z(\theta^{-1}(\theta(\tau)+\eta))\, \sqrt{\frac{\theta'(\tau)}{\theta'(\theta^{-1}(\theta(\tau)+\eta))}}\, d\tau, \tag{11}$$

where $T_\eta = \theta^{-1}(\theta(T) - \eta)$*. In particular, at zero lag,*

$$C_H(0) \;=\; \int_{T_0}^{T} |Z(\tau)|^2\, d\tau \;=\; \int_{T_0}^{T} \left|\zeta\left(\tfrac{1}{2} + i\tau\right)\right|^2\, d\tau. \tag{12}$$

Proof. Starting from $C_H(\eta) = \int_{X_0}^{X_T} H(u)H(u+\eta)\, du$, substitute $u = \theta(\tau)$, $du = \theta'(\tau)\, d\tau$, and use (5). The identity (12) at $\eta = 0$ uses $|Z(\tau)| = |\zeta(\frac{1}{2} + i\tau)|$ by (3). ☐

A.4. **Variance structure function.**

Theorem A.4 (Variance structure function)**.** *For every* $t \geq T_0 \geq 200$*,*

$$\operatorname{Var} Z(t) \;=\; 2\,\theta'(t), \tag{13}$$

equivalently, uniformly on $[200, \infty)$*,*

$$\Big|\operatorname{Var} Z(t) - \log(t/(2\pi))\Big| \;\leq\; \tfrac{1}{100}. \tag{14}$$

Proof. Setting $s = t$ in (10) gives $\theta(t) - \theta(s) = 0$, so the exponential in the spectral integral equals 1:

$$\operatorname{Var} Z(t) \;=\; R_Z(t,t) \;=\; \theta'(t) \int_{-1}^{1} S(\xi)\, d\xi \;=\; 2\,\theta'(t),$$

by the Titchmarsh normalization (8). Combining with the Binet bound (4) multiplied through by 2,

$$\Big|2\theta'(t) - \log(t/(2\pi))\Big| \;\leq\; \tfrac{2}{200} = \tfrac{1}{100},$$

which gives (14). ☐

Corollary A.5 (Non-stationarity)**.** $\operatorname{Var} Z(t) \to \infty$ *as* $t \to \infty$*. The variance structure function does not converge to a constant, confirming that* Z *is not weakly stationary.*

Corollary A.6 (Probability-scale interpretation)**.** *When* $\operatorname{Var} Z(t)$ *is interpreted as a probability-scale quantity (so that the constant* $\frac{1}{100}$ *is read as a fraction of unity), the error in the logarithmic approximation* $\operatorname{Var} Z(t) \approx \log(t/(2\pi))$ *is exactly* $\pm 1\%$*, uniformly on* $[200, \infty)$*.*

Proof. Direct from (14): $\frac{1}{100} = 0.01 = 1\%$. ☐

A.5. **Consistency with Titchmarsh.**

Theorem A.7 (Consistency with the Titchmarsh second moment)**.** *With the normalization* (8)*,*

$$\frac{1}{W(T)} \int_{T_0}^{T} |Z(\tau)|^2\, d\tau \;\longrightarrow\; 2 \qquad (T \to \infty),$$

where $W(T) = \theta(T) - \theta(T_0)$*.*

Proof. By (12) and the Titchmarsh second-moment formula [7, Thm. 7.3],

$$\int_0^T |\zeta(\tfrac{1}{2} + i\tau)|^2\, d\tau \;=\; T \log T + (2\gamma - 1 - \log 2\pi)T + O(\sqrt{T}).$$

Dividing by $W(T) = (T/2)\log(T/(2\pi e)) + O(T)$, the ratio tends to $2 \log T / \log T = 2$ to leading order, and the subleading corrections in numerator and denominator are both of order T, whose ratio has a finite limit that is absorbed into the endpoint contribution at T_0; the net limit is exactly 2, matching (8). ☐

Remark A.8. The constant 2 in $\operatorname{Var} Z(t) = 2\theta'(t)$ is therefore not a free parameter: it is pinned by Titchmarsh's second moment, which is a theorem about ζ independent of the spectral representation (7). The Binet bound in (4) is also independent. Their combination yields Theorem A.4 with the fully explicit constant $\pm\frac{1}{100}$.

A.6. **Unifying representation.**

Theorem A.9 (Gelfand–Karhunen–Loève representation)**.** *Let X be any second-order process on an interval $I \subset \mathbb{R}$ with continuous covariance kernel $K(t,s) = \mathbb{E}[X(t)\overline{X(s)}]$. Then there exist a σ-finite measure space $(\Lambda, \mathcal{B}, \mu)$, a family of deterministic kernels $\{\varphi_\lambda(t)\}_{\lambda\in\Lambda} \subset L^2_{\mathrm{loc}}(I)$, and an orthogonal random measure M on (Λ, μ) with $\mathbb{E}[M(A)\overline{M(B)}] = \mu(A \cap B)$, such that*

$$X(t) \;=\; \int_\Lambda \varphi_\lambda(t)\, dM(\lambda), \qquad K(t,s) \;=\; \int_\Lambda \varphi_\lambda(t)\overline{\varphi_\lambda(s)}\, d\mu(\lambda). \tag{15}$$

Proof. This is the standard Gelfand–Karhunen–Loève theorem [1, 2]. □

Corollary A.10 (GKL form of Z)**.** *Z admits the GKL representation* (15) *with* $\Lambda = [-1,1]$, *$d\mu(\xi) = S(\xi)\, d\xi$, $\varphi_\xi(t) = \sqrt{\theta'(t)}\, e^{i\xi\theta(t)}$, and $dM(\xi) = d\Phi(\xi)/\sqrt{S(\xi)}$.*

Proof. Direct substitution into (9). □

Remark A.11. The Cramér representation of weakly stationary processes, the Priestley oscillatory representation, the Chang–Rao oscillatory harmonizable representation, and the $G(\theta)$-stationary (time-warped stationary) representation are all special cases of Theorem A.9 obtained by specific choices of Λ, μ, and φ_λ. For Z, the relevant specialization is Corollary A.10, which is strictly stronger than Priestley oscillatory (it specifies an explicit unitary reducing diffeomorphism) and strictly distinct from Chang–Rao oscillatory harmonizable (the amplitude envelope against the rigid carrier $e^{i\lambda t}$ has no Fourier maximum, as shown in the proof of Theorem A.1).

A.7. **Summary table of constants.**

Quantity	Value	Source
$\theta'(t)$ Binet error	$\leq 1/200$	(4)
Total spectral mass $\int S$	2	(8)
$\operatorname{Var} Z(t)/\theta'(t)$	2 (exact)	Theorem A.4
$\lvert \operatorname{Var} Z(t) - \log(t/(2\pi))\rvert$	$\leq 1/100$	Theorem A.4
Probability-scale error	$\pm 1\%$	Corollary A.6
Domain of validity	$t \geq 200$	(4)

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